2026 Spring

Course ID: 1142-1764

Foundations of Computer Science

04-Computer networks

1

HOUR

10:10 – 11:00

Topic

Course Introduction

Cloud Computing

Course Introduction

Welcome to the course — let's get started.

Week 1

50 minutes

Course Info

Introduction to Computer Science

Fundamentals · Organization · Networks · OS · Algorithms · Data Structures · ADT

2 Semesters · 11 Topics · 32 Weeks · 3 Credits each

INSTRUCTOR	DEPARTMENT	LANGUAGE	TOOLS
Prof. H.T. Chu	**International Program in Computer Science**	**English (EMI)**	**AWS Academy**

ASSESSMENT (EACH SEMESTER)

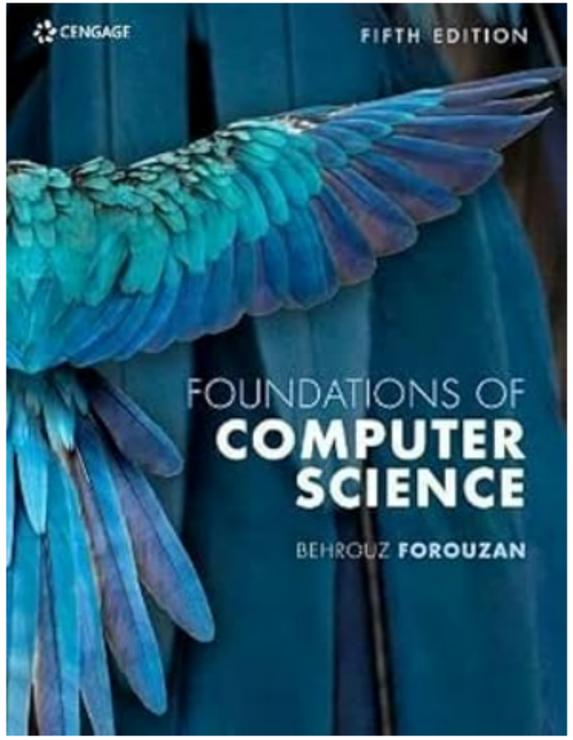
Assignments	Midterm Exam	Project / Labs	Final Exam	Participation
30%	**20%**	**30%**	**20%**	**Bonus 5%**

Providence University · Department of Computer Science and Information Engineering · AY 2025–2026

Thursday, 3-4 (10:10-12:00)

Providence Hall R307

The Textbook



Foundations of Computer Science Paperback

作者 Behrouz Forouzan (Author)

The symposium was held Oct.-Nov. 1989, Research Triangle Park, North Carolina. One hundred papers in theoretical computer science treat dispersers, deterministic amplification, and weak random sources; efficient NC algorithms for set cover with applications to learning and geometry; the inverse of automorphism in polynomial time; and speeding-up linear programming using fast matrix multiplication. Acidic paper; no subject index. Annotation copyright Book News, Inc. Portland, Or.

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語言	尺寸	ISBN-10	ISBN-13
English	18.9 x 24.7 x 2.6 cm	1473787327	978-1473787322

查看所有詳細內容

1 Introduction

Turing Model
Von Neumann Model
Computer Components
History
Social and Ethical Issues
Computer Science as a Discipline
Outline of the course
End-Chapter Materials
Practice Set

2 Number Systems

Introduction
Positional Number Systems
Non-Positional Number Systems
End-Chapter Materials
Practice Set

3 Data Storage

Data Types
Storing Numbers
Storing Text
Storing Audio
Storing Images
Storing Video
End-Chapter Materials
Practice Set

4 Operations on Data

Logic Operations
Shift Operations
Arithmetic Operations
End-Chapter Materials
Practice Set

5 Computer Organization

Introduction
Central Processing Unit
Main Memory
Input/Output Subsystem
Subsystem Interconnection
Program Execution
Different Architectures
A Simple Computer
End-Chapter Materials
Practice Set

6 Computer Networks

Introduction
TCP/IP Protocol Suite
Layers
Internet Applications
The World Wide Web (www)
End-Chapter Materials
Practice Set

7 Operating Systems

Introduction
Evolution
Components
A Survey of Operating Systems
End-Chapter Materials
Practice Set

8 Algorithms

Concept
Three constructs
Algorithm Representation
A More Formal Definition
Basic Algorithms
Subalgorithms
Recursion
End-Chapter Materials
Practice Set

9 Efficiency of Algorithms

Definition
Common Algorithms
Big-O Notation
End-Chapter Materials
Practice Set

10 Programming Languages

Evolution
Translation
Programming Paradigms
Common Concepts
End-Chapter Materials
Practice Set

11 Software Engineering

The Software Life cycle
Analysis Phase
Design Phase
Implementation Phase
Testing Phase
Documentation
End-Chapter Materials
Practice Set

12 Data Structure

Arrays
Records
Linked Lists
End-Chapter Materials
Practice Set

13 Abstract Data Types

Background
Stacks
Queues
General Linear Lists
Trees
Graphs
End-Chapter Materials
Practice Set

This semester

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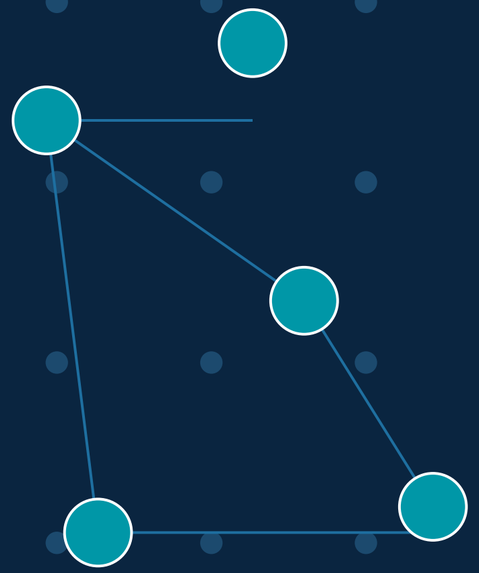
FUNDAMENTALS OF COMPUTER SCIENCE

Introduction to

Computer Networks

- How computers communicate
- Network types, protocols & security
- The Internet & beyond



Course Outline

01 **What is a Network?**
Definitions, history, importance

02 **Network Types & Topology**
LAN, WAN, MAN, PAN; Bus, Star, Mesh…

03 **Network Hardware**
NIC, Switch, Router, Modem, AP

04 **OSI & TCP/IP Models**
7-layer model, protocols

05 **Addressing & Routing**
IP, IPv4/v6, DNS, DHCP

06 **Application Protocols**
HTTP, FTP, Email, HTTPS

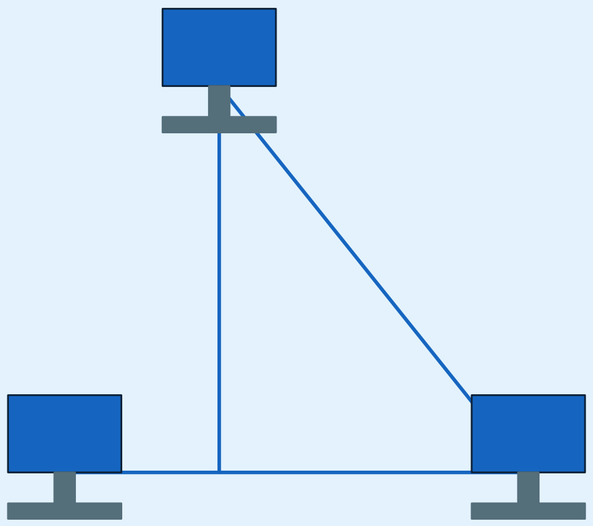
07 **Wireless & Security**
Wi-Fi, Firewall, VPN, Encryption

08 **The Internet & Future**
Cloud, IoT, 5G

What is a Computer Network?

A computer network is a group of two or more computers (or devices) connected together to share resources, exchange data, and communicate.

- Share files, printers, and internet connections
- Enable communication (email, messaging, video calls)
- Centralize data storage and management
- Support distributed applications and services



Computer Network

Connected devices sharing resources

Brief History of Computer Networks

1960s

ARPANET

First packet-switching network; precursor to the Internet

1970s

TCP/IP

Vint Cerf & Bob Kahn design the foundational protocol suite

1980s

Ethernet & LAN

Local Area Networks become common in offices

1990s

World Wide Web

Tim Berners-Lee invents the Web; public Internet explodes

2000s

Wi-Fi & Mobile

Wireless networks, smartphones, social media

2010s+

Cloud & IoT

Billions of devices, edge computing, 5G networks

Why Do Networks Matter?

5.4B+

Internet users worldwide (2024)

15B+

Connected IoT devices globally

4.9%

Annual network traffic growth rate

99%

Fortune 500 firms rely on cloud networks

Networks power everything — from browsing the web, sending emails, streaming videos, to running entire businesses and hospitals.

Understanding networks is essential for every computer scientist!

SECTION 02

Network Types & Topology

PAN · LAN · MAN · WAN · Bus · Star · Ring · Mesh

Types of Networks by Scale

PAN	LAN	MAN	WAN
Personal Area Network	Local Area Network	Metropolitan Area Net.	Wide Area Network
Range	Range	Range	Range
~10m	~100m–1km	~10–100km	1000s km
Example	Example	Example	Example
Bluetooth headset, smartwatch	Home, office, school campus	City-wide ISP, cable TV	The Internet, corporate WANs

What is Network Topology?

Network topology describes the arrangement of nodes and connections in a network — both physical (cables/hardware) and logical (data flow).

Bus Topology

Single shared line

Star Topology

Central hub/switch

Ring Topology

Circular closed loop

Mesh Topology

Every node connected

Tree Topology

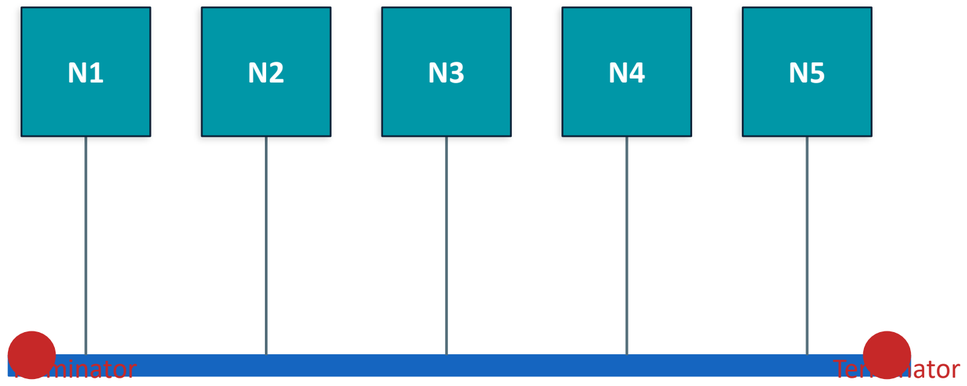
Hierarchical layers

Hybrid Topology

Combination of types

Bus Topology

- All devices connect to a single backbone cable
- Data travels along the bus and is received by all devices
- Simple and inexpensive to implement
- A break in the cable disables the entire network
- Not scalable — performance degrades with more nodes

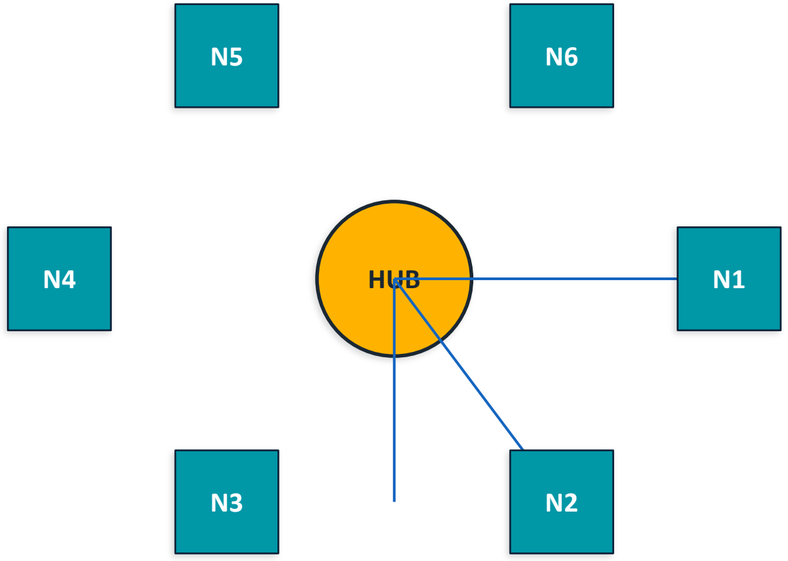


Bus Topology

✗ Single point of failure

Star Topology

- All devices connect to a central hub or switch
- The hub/switch manages all network traffic
- Most common in modern Ethernet LANs
- Failure of one node doesn't affect others
- Central hub failure brings down entire network

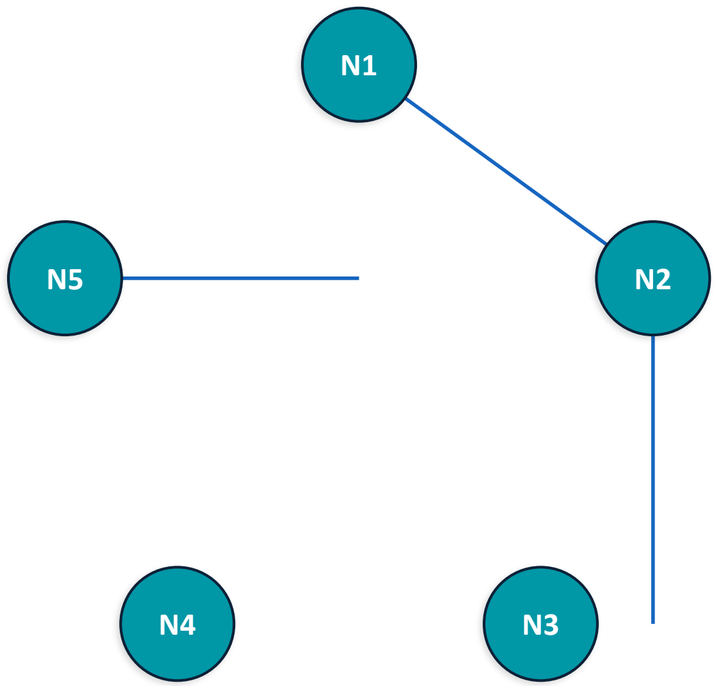


Star Topology

★ **Most widely used topology today — easy to manage and troubleshoot**

Ring Topology

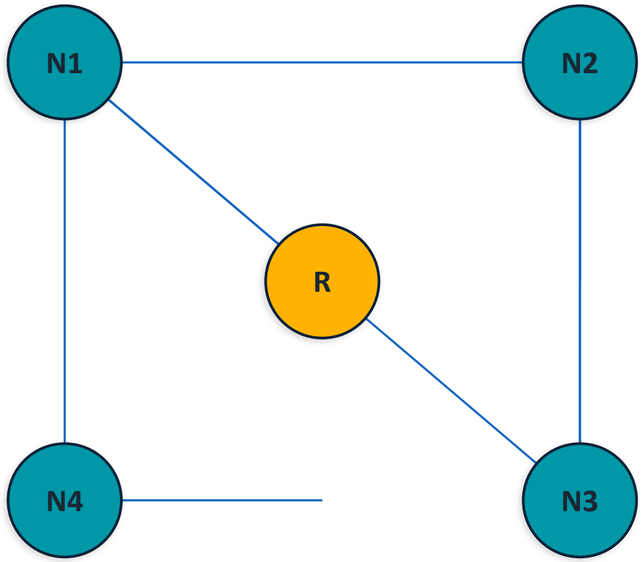
- Devices connected in a closed loop
- Data travels in one (or both) directions around the ring
- Each node acts as a repeater
- Used in older token ring networks (IEEE 802.5)
- One failure can disrupt the entire ring (unless dual-ring)



Ring Topology

Mesh Topology

- Every device connects to every other device
- Full mesh: n(n-1)/2 connections for n nodes
- Extremely fault-tolerant — multiple paths available
- Used in military, critical infrastructure, and the Internet backbone
- Expensive and complex to implement for large networks



Mesh Topology

(Full mesh — 10 links for 5 nodes)

SECTION 03

Hardware

NIC · Hub · Switch · Router · Modem · Access Point

Network Interface Card (NIC)

- Hardware component allowing a device to connect to a network
- Each NIC has a unique 48-bit MAC address (e.g., A1:B2:C3:D4:E5:F6)
- Wired NICs use Ethernet (RJ-45 connector)
- Wireless NICs support Wi-Fi (802.11a/b/g/n/ac/ax)
- Speeds: 10 Mbps → 100 Mbps → 1 Gbps → 10 Gbps

MAC Address

A1 : B2 : C3 : D4 : E5 : F6

6 bytes (48 bits) — burned into hardware

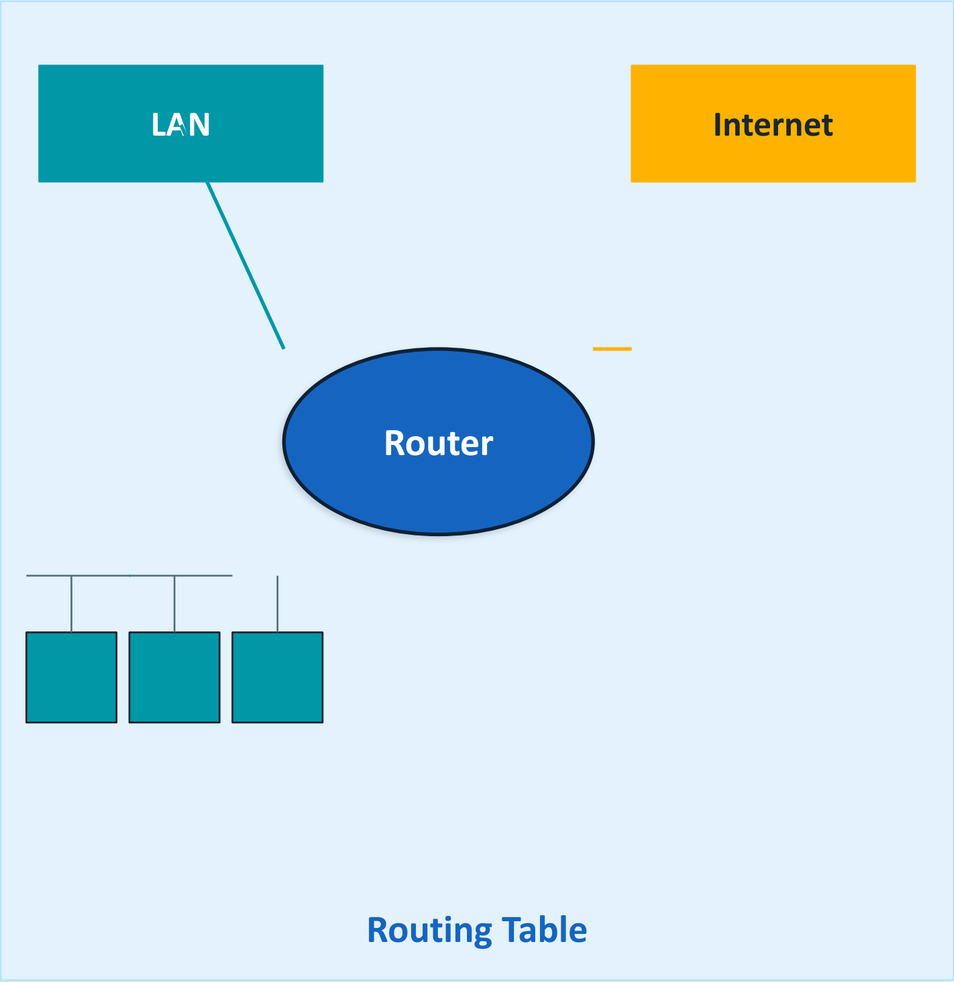
Wired NIC	Wireless NIC	Virtual NIC
Ethernet port, RJ-45	Wi-Fi, 802.11ax	Software-defined

Hub vs. Switch

HUB		SWITCH ★
• Layer 1 device (Physical) • Broadcasts data to ALL ports • Creates one collision domain • Half-duplex communication • Slower, less efficient • Mostly obsolete today	VS	• Layer 2 device (Data Link) • Sends data only to target port • Each port = own collision domain • Full-duplex communication • Fast, efficient switching • Standard in modern networks

Router

- Layer 3 (Network) device that routes packets between networks
- Uses IP addresses to determine the best path for data
- Connects your home LAN to the Internet (WAN)
- Maintains routing tables for path selection
- Protocols: RIP, OSPF, BGP (dynamic routing)



Modem & Wireless Access Point

MODEM

MOdulator / DEModulator

- Converts digital ↔ analog signals
- Connects to ISP over telephone or cable
- DSL, Cable, Fiber types
- Enables home Internet access

+

ACCESS POINT (AP)

Wireless Network Bridge

- Connects wireless devices to wired LAN
- Broadcasts SSID (network name)
- Supports multiple devices simultaneously
- Standards: 802.11a/b/g/n/ac/ax (Wi-Fi 6)

SECTION 04

OSI & TCP/IP Models

Standardized frameworks for network communication

The OSI Reference Model

7	Application	User-facing services: HTTP, FTP, DNS, SMTP
6	Presentation	Data format, encryption, compression
5	Session	Session setup, maintenance, teardown
4	Transport	End-to-end delivery: TCP, UDP; flow control
3	Network	Routing & addressing: IP, ICMP
2	Data Link	Framing, MAC addressing, error detection
1	Physical	Bits over cables/fiber/radio; voltages & signals

Please Do Not Throw Sausage Pizza Away (mnemonic)

Layers 1 & 2: Physical & Data Link

Layer 1 — Physical

- Transmission of raw bits over medium
- Media types: copper, fiber, radio, coax
- Connectors: RJ-45, BNC, SC fiber
- Specifications: 10BASE-T, 100BASE-TX
- Devices: Repeater, Hub, Cable

Layer 2 — Data Link

- Organizes bits into frames
- MAC addressing for local delivery
- Sub-layers: LLC (IEEE 802.2) & MAC
- Error detection (CRC checksum)
- Devices: Switch, Bridge, NIC

Layers 3 & 4: Network & Transport

Layer 3 — Network

- Logical addressing (IP addresses)
- Routing packets across networks
- Protocols: IP, ICMP, IGMP, ARP
- Devices: Router, Layer-3 Switch
- Fragmentation and reassembly

Layer 4 — Transport

- End-to-end reliable delivery
- Port numbers (0–65535)
- TCP: connection-oriented, reliable
- UDP: connectionless, fast
- Flow control & congestion control

TCP vs. UDP

Feature	TCP	UDP
Connection	Connection-oriented	Connectionless
Reliability	Guaranteed delivery	No delivery guarantee
Order	In-order delivery	No ordering
Speed	Slower (overhead)	Faster (lightweight)
Use case	HTTP, FTP, Email	Video stream, DNS, VoIP
Header size	20–60 bytes	8 bytes

Layer 5–7: Session, Presentation & Application

The upper three layers are often treated together in practice, handling session management, data formatting, and user-facing services.

HTTP/HTTPS

Web browsing

FTP/SFTP

File transfer

SMTP/IMAP

Email services

DNS

Name resolution

SSH

Secure shell

SSL/TLS

Encryption/presentation

TCP/IP Model

The TCP/IP model is the practical framework used by the Internet. It simplifies the OSI model into 4 layers.

TCP/IP Layer	Protocols	OSI Mapping
Application	HTTP, FTP, DNS, SMTP, SSH	OSI Layers 7-5
Transport	TCP, UDP	OSI Layer 4
Internet	IP, ICMP, ARP	OSI Layer 3
Link	Ethernet, Wi-Fi, Physical media	OSI Layers 2-1

SECTION 05

Addressing & Routing

IP Addresses · IPv4 · IPv6 · DNS · DHCP · Subnetting

IPv4 Addressing

192 . 168 . 1 . 100

Octet 1	Octet 2	Octet 3	Octet 4
11000000	10101000	00000001	01100100

32-bit address

~4.3 billion unique addresses

Dotted-decimal

Four octets (0–255 each)

Classes

A (1–126), B (128–191), C (192–223)

Private ranges

10.x.x.x, 172.16–31.x.x, 192.168.x.x

IPv4 vs. IPv6

Feature	IPv4	IPv6
Address size	32 bits	128 bits
Format	Dotted decimal (192.168.1.1)	Hexadecimal (2001:db8::1)
Total addresses	~4.3 billion	~3.4×10^{38}
Header size	20–60 bytes (variable)	40 bytes (fixed)
NAT required	Yes (address exhaustion)	No (vast address space)
Security	IPSec optional	IPSec built-in
Auto-config	DHCP required	SLAAC — stateless auto-config

Subnetting Basics

Subnetting divides a large network into smaller sub-networks (subnets) for better performance and security.

```
192.168.1.0  /  24
```

← Network Address → ← Prefix Length →

CIDR	Subnet Mask	Hosts	Use
/8	255.0.0.0	16,777,216	Class A
/16	255.255.0.0	65,536	Class B
/24	255.255.255.0	256	Class C
/30	255.255.255.252	4	Point-to-point

Domain Name System (DNS)

DNS translates human-readable domain names (www.example.com) into IP addresses (93.184.216.34) — the Internet's phone book.

1 User types URL
Browser checks local cache

→

2 Recursive Resolver
ISP's DNS resolver queries further

→

3 Root Name Server
Directs to TLD server (.com, .org)

→

4 TLD Name Server
Directs to authoritative name server

→

5 Authoritative DNS
Returns the actual IP address

DNS uses Port 53 (UDP for queries, TCP for zone transfers) · TTL controls cache duration

DHCP — Dynamic Host Configuration Protocol

DHCP automatically assigns IP addresses and network configuration to devices on a network, eliminating manual setup.

D — **DISCOVER**

Client broadcasts: 'Is there a DHCP server?'

O — **OFFER**

Server responds with available IP address

R — **REQUEST**

Client requests the offered IP address

A — **ACK**

Server confirms and leases the IP address

SECTION 06

Application Protocols

HTTP · HTTPS · FTP · SMTP · POP3 · IMAP

HTTP & HTTPS

HTTP (Port 80)

- HyperText Transfer Protocol
- Stateless request-response model
- Methods: GET, POST, PUT, DELETE
- Status codes: 200 OK, 404 Not Found, 500 Error
- No encryption — data sent in plain text

+S

HTTPS (Port 443) 🔒

- HTTP Secure = HTTP + TLS/SSL
- Encrypts data in transit
- Provides authentication via certificates
- Required for logins, payments, and privacy
- Indicated by padlock in browser address bar

File Transfer Protocol (FTP)

- Protocol for transferring files between client and server
- Uses two separate connections: Port 21 (control) and Port 20 (data)
- Active mode: server initiates data connection back to client
- Passive mode: client initiates both control and data connections
- FTP transmits credentials in plain text — use SFTP or FTPS for security

FTP	**Port 20/21** No encryption
FTPS	**Port 990** TLS/SSL secured
SFTP	**Port 22** SSH-based, fully encrypted

Email Protocols: SMTP, POP3 & IMAP

SMTP — Port 25 / 587

SENDING email

- Simple Mail Transfer Protocol
- Used by mail clients to send email to server
- Also used for server-to-server relay
- STARTTLS for encryption on port 587

POP3 — Port 110 / 995

RECEIVING email

- Post Office Protocol v3
- Downloads email from server to client
- Usually deletes from server after download
- Good for single-device access

IMAP — Port 143 / 993

SYNC email

- Internet Message Access Protocol
- Keeps email on the server
- Sync across multiple devices
- Preferred for modern multi-device use

SECTION 07

Wireless & Security

Wi-Fi · Firewalls · Encryption · VPN · Threats

Wireless Networking (Wi-Fi)

Standard	Year	Band	Max Speed	Wi-Fi Name
802.11b	1999	2.4 GHz	11 Mbps	546E7A
802.11g	2003	2.4 GHz	54 Mbps	1565C0
802.11n	2009	2.4/5 GHz	600 Mbps	0097A7
802.11ac	2013	5 GHz	3.5 Gbps	FFB300
802.11ax	2019	2.4/5/6 GHz	9.6 Gbps	**2E7D32**

Security: WEP (broken) → WPA → WPA2 (AES) → WPA3 (latest standard, strongest protection)

Network Security Fundamentals

Common Threats

Malware

Viruses, worms, ransomware attacking systems

Phishing

Deceptive emails/sites stealing credentials

DDoS

Overwhelming servers with massive traffic

Man-in-Middle

Intercepting communication between parties

Defenses

Firewall

Filters traffic based on security rules

IDS/IPS

Detects and prevents intrusions in real-time

Encryption

Protects data confidentiality with crypto

Authentication

Verifies identity with passwords, MFA, certs

Firewalls, VPN & Encryption

Firewall

- Monitors and controls incoming/outgoing traffic
- Rules based on IP, port, protocol
- Types: Packet-filter, Stateful, Application-layer
- First line of defense in network security

VPN (Virtual Private Network)

- Creates an encrypted tunnel over public Internet
- Hides user IP and location
- Protocols: OpenVPN, IPSec, WireGuard, SSL VPN
- Used for remote work and privacy

Encryption

- Transforms data so only authorized parties can read it
- Symmetric: AES-256 (same key both sides)
- Asymmetric: RSA, ECC (public + private key pair)
- TLS/HTTPS encrypts web communications

SECTION 08

The Internet

& The Future

Cloud Computing · IoT · 5G · Edge Computing · SDN

Summary & The Future of Networking

Networks

Connect the world via LAN, WAN, the Internet

Topology

Bus, Star, Ring, Mesh — each with trade-offs

OSI/TCP/IP

7-layer models guide protocol design

Addressing

IP, DNS, DHCP make the Internet work

Security

Firewalls, VPN, encryption protect data

Future Trends:

5G Networks · Edge Computing · IoT (50B+ devices) · Software-Defined Networking · Quantum Networking

SECTION 09

Networking in AWS Cloud

VPC · Subnets · Route 53 · Load Balancing · CloudFront · Direct Connect · VPN

What is AWS? Cloud Computing Overview

IaaS

Infrastructure as a Service

Compute, Storage, Networking — EC2, S3, VPC

PaaS

Platform as a Service

Deploy apps without managing servers — Elastic Beanstalk, RDS

SaaS

Software as a Service

Fully managed apps — Amazon WorkMail, Chime

Shared Responsibility Model

AWS manages: Physical hardware, data centers, hypervisor, managed services

AWS Security
of the Cloud

You manage: OS, firewall rules, app code, data, IAM policies, network config

Your Security
in the Cloud

AWS Global Infrastructure

33+

Regions

Geographic areas, each fully independent (e.g., us-east-1, ap-southeast-1). Choose based on latency, compliance, and cost.

105+

Availability Zones

Isolated data centers within a region. Each AZ has independent power, cooling, and networking.

600+

Edge Locations

CloudFront CDN PoPs for caching content close to users. Reduces latency for global audiences.

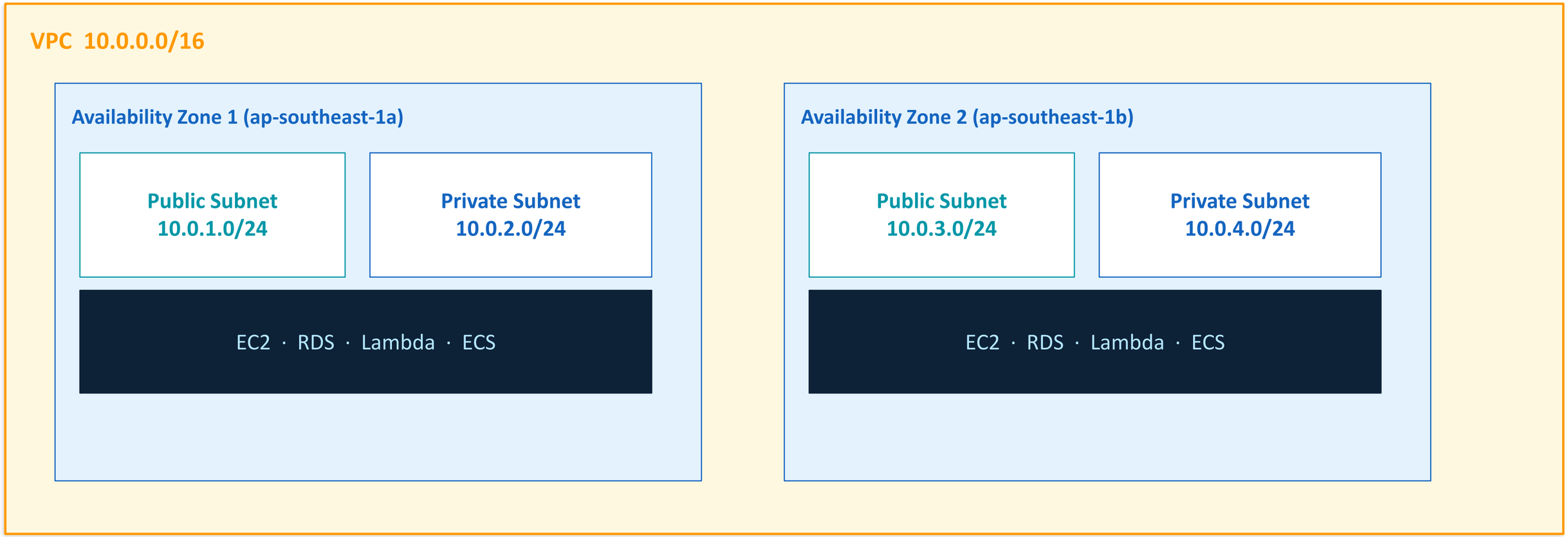
30+

Local Zones

AWS infrastructure closer to large population centers for single-digit ms latency workloads.

Amazon VPC — Virtual Private Cloud

A VPC is your own logically isolated virtual network within AWS. You have complete control over IP ranges, subnets, routing, and security.



VPC Core Components

Subnets

Logical segments of a VPC's IP space. Public subnets route to IGW; private subnets don't.

Route Tables

Define rules for routing traffic out of a subnet. Each subnet must be associated with one route table.

Internet Gateway

Enables instances in public subnets to communicate with the Internet. Horizontally scaled and HA.

NAT Gateway

Allows private subnet instances to access the Internet outbound while remaining unreachable inbound.

Security Groups

Stateful virtual firewall at the instance level. Allow rules only; denies are implicit.

NACLs

Stateless subnet-level firewall. Support both Allow and Deny rules; evaluated in order.

Security Groups vs. Network ACLs

Feature	Security Group	Network ACL (NACL)
Scope	Instance / ENI level	Subnet level
Statefulness	Stateful — return traffic auto allowed	Stateless — return traffic needs explicit rule
Rule types	Allow rules only	Allow AND Deny rules
Rule evaluation	All rules evaluated together	Rules evaluated in numbered order (lowest first)
Default behavior	Deny all inbound; allow all outbound	Allow all inbound and outbound
Use case	Fine-grained instance protection	Broad subnet-level traffic control

Internet Gateway & NAT Gateway

Internet Gateway (IGW)

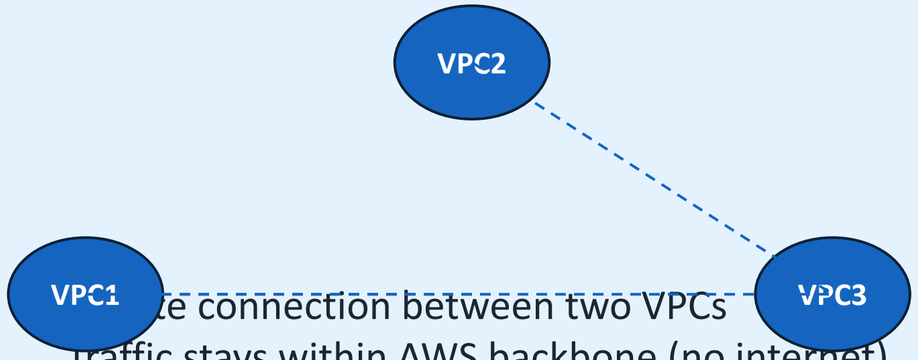
- Attached to a VPC (one per VPC)
- Enables bidirectional Internet access
- Instances need public IP or Elastic IP
- Route table entry: 0.0.0.0/0 → igw-xxxxx
- Horizontally scaled, HA, no bandwidth limit
- Free — only pay for data transfer

NAT Gateway

- Placed in a public subnet
- Private instances → Internet (outbound only)
- Hides private IPs behind Elastic IP
- Route: 0.0.0.0/0 → nat-xxxxx (in private RT)
- Managed by AWS — highly available per AZ
- Charged per hour + per GB processed

VPC Peering & AWS Transit Gateway

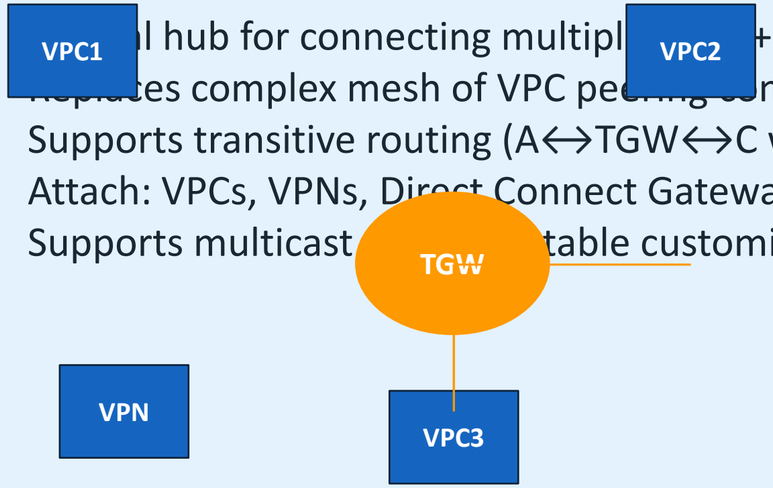
VPC Peering



- [illegible]te connection between two VPCs
- Traffic stays within AWS backbone (no internet)
- Can peer cross-account and cross-region
- Non-transitive — A↔B and B↔C does NOT mean A↔C
- No overlapping CIDR blocks allowed

Transit Gateway (TGW)

- [illegible]l hub for connecting multipl[illegible] + on-prem
- [illegible]ces complex mesh of VPC pe[illegible]nnections
- Supports transitive routing (A↔TGW↔C works!)
- Attach: VPCs, VPNs, Direct Connect Gateways
- Supports multicast [illegible]table customization



Elastic Load Balancing (ELB)

ELB automatically distributes incoming traffic across multiple targets (EC2, containers, IPs) for high availability and fault tolerance.

Application Load Balancer (ALB)

Layer 7

- HTTP/HTTPS routing
- Path & host-based rules
- WebSocket & HTTP/2
- Best for web apps & microservices

Network Load Balancer (NLB)

Layer 4

- TCP/UDP/TLS routing
- Ultra-low latency
- Handles millions req/sec
- Static IP per AZ support

Gateway Load Balancer (GWLB)

Layer 3

- Deploys virtual appliances
- Inline traffic inspection
- Third-party firewalls/IDS
- Transparent bump-in-wire

Amazon Route 53 — Cloud DNS

Route 53 is AWS's highly available and scalable DNS web service. It also handles domain registration and health checking.

Routing Policies

Simple

Single resource; no health check

Weighted

Split traffic by percentage

Latency-based

Route to lowest-latency region

Failover

Active/passive with health checks

Geolocation

Route by user's geographic location

Geoproximity

Route by distance with bias control

Multi-value

Returns multiple IPs for availability

IP-based

Route based on client IP address

Amazon CloudFront — Content Delivery Network

CloudFront is a global CDN that delivers content (web pages, videos, APIs) from 600+ Edge Locations close to end users with low latency.

HTTPS / TLS

Free SSL certs via ACM, HTTP→HTTPS redirect

Origin Shield

Additional caching layer to reduce origin load

Lambda@Edge

Run code at edge — auth, redirects, A/B testing

WAF Integration

Block DDoS, SQL injection, XSS at the edge

AWS VPN — Site-to-Site & Client VPN

Site-to-Site VPN

- Encrypted IPSec tunnel between on-premises and VPC
- Virtual Private Gateway (VGW) attached to VPC
- Customer Gateway (CGW) represents your on-prem device
- Supports static or dynamic (BGP) routing
- Two tunnels per connection for redundancy
- Bandwidth up to ~1.25 Gbps; use TGW for more

AWS Client VPN

- OpenVPN-based managed VPN for remote users
- Users connect to VPC using an OpenVPN client
- Integrates with Active Directory for auth
- Split-tunnel or full-tunnel options
- Scales automatically — no server management
- Supports MFA with SAML, ACM certificates

AWS Direct Connect

Direct Connect provides a dedicated private network connection from your on-premises data center to AWS — bypassing the public Internet.

Speed Options:	1 Gbps, 10 Gbps, 100 Gbps (hosted: 50 Mbps – 10 Gbps)
Consistency:	Consistent network performance — no Internet variability
Cost Savings:	Lower data transfer costs vs Internet; no egress charges on DX
Virtual Interfaces:	Public VIF (AWS services), Private VIF (VPC), Transit VIF (TGW)

VPC Endpoints & AWS PrivateLink

Connect your VPC to AWS services and third-party services privately — without traversing the public Internet.

Gateway Endpoint

- Free — no hourly charge
- Supports S3 and DynamoDB only
- Added as route table entry
- Traffic stays within AWS network
- Regional — cannot cross regions

Interface Endpoint (PrivateLink)

- ENI with private IP in your subnet
- Supports 100+ AWS services
- Works with DNS — resolve to private IP
- Cross-account & cross-region support
- Hourly + data processing charge

Gateway Load Balancer Endpoint

- Intercept and inspect traffic
- Third-party virtual appliances
- Inline bump-in-wire architecture
- Used with GWLB for security services
- East-west and north-south traffic

AWS Network Monitoring & Visibility

VPC Flow Logs

Capture metadata about IP traffic going to/from ENIs. Published to CloudWatch Logs or S3. Useful for troubleshooting and security audits. Does NOT capture packet content.

AWS CloudWatch

Metrics, alarms, and dashboards for network resources. Monitor ELB request counts, latency, NAT gateway bytes, and more. Set alarms for threshold breaches.

AWS Network Manager

Centrally manage and monitor global network built on AWS. Visualize topology, track events, and monitor performance across TGW and SD-WAN integrations.

VPC Reachability Analyzer

Check connectivity between two endpoints. Without sending actual traffic. Identify misconfigured security groups, NACLs, or route tables blocking traffic.

AWS X-Ray

Distributed tracing for application-level network calls. Trace requests end-to-end across microservices, identify latency hotspots and errors.

Traffic Mirroring

Copy network traffic from ENIs to monitoring appliances or collectors. Deep packet inspection, threat detection, and network forensics without disrupting workloads.

AWS WAF, Shield & Network Firewall

AWS WAF

Web Application Firewall

- Filters HTTP/S at Layer 7
- Block SQL injection, XSS, CSRF
- Managed rule groups (OWASP Top 10)
- Custom rules with IP sets & rate limits
- Integrates with CloudFront, ALB, API GW

AWS Shield

DDoS Protection

- Shield Standard: FREE for all AWS customers
- Automatic protection from common DDoS attacks
- Shield Advanced: $3,000/mo + 24/7 DRT support
- Near-real-time attack visibility
- Cost protection against DDoS-related scaling

AWS Network Firewall

Managed VPC Firewall

- Stateful and stateless rule groups
- Deep packet inspection (Suricata rules)
- Centrally manage across accounts (Firewall Manager)
- TLS inspection capability
- Deploy in each VPC or centrally via TGW

Common AWS Network Architecture Patterns

3-Tier Web App

Public subnet (ALB) → Private subnet (App servers) → Isolated subnet (RDS). Classic pattern for web workloads.

Hub-and-Spoke

Central shared-services VPC connected to spoke VPCs via TGW or peering. Centralizes logging, security, and egress.

Hybrid Cloud

On-premises connected to AWS via Direct Connect or VPN. Extends corporate network into the cloud securely.

Multi-Region HA

Active-active deployment across AWS regions with Route 53 latency/failover routing and Global Accelerator.

Zero Trust

No implicit trust. Every request authenticated. VPC Lattice, PrivateLink, IAM auth for service-to-service communication.

Microservices Mesh

Service mesh using AWS App Mesh, ECS, or EKS with internal ALB/NLB for east-west service discovery and routing.

On-Premises Networking vs. AWS Networking

Concept	On-Premises	AWS Equivalent
Network segment	VLAN / subnet	VPC Subnet
Firewall (host)	Host-based firewall / iptables	Security Group
Firewall (network)	ACL / perimeter firewall	Network ACL / AWS Network Firewall
Router	Physical router / L3 switch	Route Table + TGW
Load balancer	F5 BIG-IP / HAProxy / Nginx	ALB / NLB / GWLB
DNS	BIND / Active Directory DNS	Route 53 / Route 53 Resolver
VPN	IPSec appliance / firewall VPN	Site-to-Site VPN / Client VPN
Dedicated WAN	MPLS, leased line	AWS Direct Connect
CDN	Akamai / Cloudflare	Amazon CloudFront

AWS Well-Architected Networking Best Practices

The AWS Well-Architected Framework provides design principles for reliable, performant, and secure cloud networks.

Operational Excellence	Security	Reliability	Performance Efficiency	Cost Optimization
• Use VPC Flow Logs & CloudWatch • Infrastructure as Code (CloudFormation) • Automate network changes with CI/CD	• Least privilege security groups • Private subnets for databases • Enable GuardDuty & Security Hub	• Multi-AZ deployments • Health checks on load balancers • Route 53 failover routing	• CloudFront for edge caching • Placement groups for low latency • Enhanced networking (ENA, SR-IOV)	• NAT Gateway per-AZ (reduce cross-AZ) • VPC endpoints for S3/DynamoDB (free) • Reserved capacity for stable workloads

AWS Networking — Key Concepts Summary

VPC

Isolated virtual network — subnets, route tables, IGW, NAT GW

Security

Security Groups (stateful) + NACLs (stateless) = defense in depth

Load Balancing

ALB (L7) / NLB (L4) / GWLB — for HA, scaling, and inspection

DNS & CDN

Route 53 for DNS & routing policies; CloudFront for edge delivery

Connectivity

VPC Peering, TGW, VPN, Direct Connect — cloud + hybrid networks

Security Tools

WAF, Shield, Network Firewall — protect from L3 to L7 threats

Explore Further: AWS Certified Cloud Practitioner · Solutions Architect Associate · Advanced Networking Specialty

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